# *ONTOLOGIES AND KNOWLEDGE GRAPHS*

***University of Jordan King Abdullah II School for Information Technology***

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**Project Title:**

Group 4- Food Recipe

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# Ontology scenario

## ontology scenario

a platform that uses an ontology to allow users to search for recipes based on their specific needs, such as the ingredients used or the type of cookware required. For example, a user wants to make a quick lunch using fish and vegetables by using a cast iron. The system suggests relevant recipes that matches the given criteria.

## ontology purpose

The purpose of the ontology is to establish a framework that define and organize various features of food recipes. this includes ingredient used, meal-type, cooking method and more. this framework facilitates efficient searching, filtering and personalized recipe recommendations.

## possible users

1.home cooks: individual who enjoy preparing meals at home and seek instructions for specific recipes.

2.professional chefs: chefs who need access to a wide range of recipes for inspiration and menu planning.

3.nutritionists and dietitians: professional who need to recommend recipes for health goals or specific dietary requirements.

4.food bloggers: individuals who create content about food and cooking and needs access to set of recipes in a structured and reusable manner.

5.software developers: developers building recipe recommendation engine, cooking apps.

## uses of the ontology

1.search and discovery: users can search for recipes based on ingredients, cooking method, cooking-time.

2.personalized recommendations: can provide personalized recipe suggestions based on user preferences and dietary needs.

3.recipe sharing: users can contribute their own recipes, share them with others.

4.Interoperability:by creating interoperable applications that enable integration of recipe data across platforms, apps, and websites.

Using an ontology helps make food recipes consistent, easy to find, and useful for all kinds of users.

# Ontology scope

1.Cooking_Method (Baking, Barbeque, Blanching, Boiling, Deep_Frying, Grilling,

Poaching, Roasting, Steaming)

**Justification:** *allow users to search for recipes based on how they are cooked.*

2.Cooking_Time (Quick_Recipe, Slow-Cooked_Recipe)

**Justification:** *help user to find recipes that fit their schedules*

3.Cookware (Aluminum, Anodized_Aluminum, Carbon_Steel, Cast_Iron, Ceramic,

Copper, Enameled_Cast_Iron, Nonstick, Stainless_Steel)

**Justification:** *important for guiding users through recipe preparation steps effectively.*

4.Ingredient (Bread_Ingredient, Dairy_Ingredient, Fat_Ingredient, Fish_Ingredient, Fruit_Ingredient, Grain_Ingredient, Meat_Ingredient, Oil_Ingredient, Poultry_Ingredient, Sweet_Ingredient, Vegetable_Ingredient)

**Justification:** *Ingredients are important to any recipe and must be standardized to make it easy to filtering, substitution and dietary analysis.*

5.Instructions

**Justification:** *to guide users through the cooking process.*

6.Meal_Type (Dinner, Breakfast, Lunch)

**Justification:** *help users to find recipes based on the time of day.*

7.Measurement_Unit (Imperial, Metric)

**Justification:** *help users to achieve perfect taste every time.*

8.Preparation_Technique (Chop, Cube, Dice, Mix)

**Justification:** *help users to understand the process.*

9.Person

**Justification:** *to get the right proportions and scale the recipe accordingly.*

10.Recipe (Dessert_Recipe, Main_Course_Recipe)

**Justification:** *help user to choose the appropriate recipe.*

11.Recipe_Complexity (Complex_Recipe, Simple_Recipe, Usual_Recipe)

**Justification:** *help users to choose recipe that match their cooking skill.*

12.Recipe_Source (Chef_Recipe, Family_Recipe, Online_Recipe)

**Justification:** *help users to ensure the recipe is reliable and good quality and give credit to the creator of the recipe.*

# Ontology knowledge sources

## Source1::

https://github.com/waleedahmed7/Recipe

Ontology/blob/main/Recipe%20Ontology.owl

2. Overview

The Recipe Ontology specifies a common, machine-readable format for capturing knowledge about

recipes, ingredients, tools, and related entities. This is also applicable for different use-cases,

including culinary education, recipe recommendation, and nutritional analysis.

Class: Recipe

Subclasses:

- MainDish
- Dessert
- Beverage

Ingredients:

- Vegetables
- Proteins
- Spices

3. Properties of Objects

These define the relationships between classes. Examples of such may include:

hasIngredient: Relates a recipe to its ingredients.

requiresTool: This property indicates the tools required to prepare a particular recipe.

hasPreparationTime: This property represents the preparation time.

4. Data Properties

These represent the attributes of the classes. Possible examples include the following:

calories : Number of calories in a dish.

cookingTime: Time needed for cooking.

difficultyLevel: A measure of how hard the recipe is to prepare.

5. Instances

If the ontology contains concrete individuals (instances), they will be described here. For example:

Instances of Ingredients:

- Tomato
- Salt
- Olive Oil

Instances of Tools:

- Blender
- Oven

6. Key Annotations

Annotations are used to provide descriptions or metadata for classes, properties or the ontology

itself. Examples:

Definitions of key terms.

Usage guidelines.

7. Use Cases :This ontology can be applied in the following domains:

Recipe Recommendation Systems: The ontology should be able to filter recipes by ingredient availability, dietary restrictions, or cuisine.

Nutritional Analysis: Recipes are analyzed for their nutrient content based on

ingredients.

Recipe Helpers: Design intelligent assistants to guide a user through any recipe.

8. Summary The Recipe Ontology provides a powerful structure for organizing knowledge in the

culinary domain: from the modeling of the relationships among recipes, ingredients, and tools up to

supporting any kind of applications, from educational tools to intelligent systems.

## Source2::

ontology Report for Recipes ;;https://github.com/acsalcedo/recipe-ontology

ws/blob/master/ontology/recipes.owl

1. Overview

recipes.owl This is an ontology that was created to model knowledge about recipes. Ontologies are

structured frameworks for organizing information, and this would contain a class, property, and the

relationships that define what a recipe constitutes-the various ingredients, ways of cooking, and

such allied concepts.

2. Key Components

2.1 Classes

Classes represent the major concepts in the ontology. From a typical recipe ontology, the following

classes may occur:

Recipe: Central concept, it defines different recipes.

Ingredient: Components utilized in the recipe.

Cooking Method: This refers to methods used in food preparation.

Cuisine: Categories according to cultural or regional styles.

Equipment: Materials needed for preparation.

Nutritional Information: Health- and dietary value-related data.

2.2 Object Properties

Object properties define relationships between classes. Examples might include:

hasIngredient: Relates a recipe to its ingredients.

usesMethod: A recipe is connected to a cooking method.

requiresEquipment: Relates recipes to necessary tools.

belongsToCuisine: Relates recipes to cuisines.

2.3 Data Properties

Data properties are attributes which may provide more information. Possible examples include:

cookingTime: Time it takes to prepare a recipe.

servingSize: Number of people a recipe serves.

Calories: Energy value of a dish.

2.4 Individuals

Individuals are specific instances of classes. Examples can be:

Recipes such as "Spaghetti Carbonara" or "Chocolate Cake."

Ingredients include "Eggs," "Flour," and "Salt."

Cooking methods like "Boiling," "Frying," and "Baking."

3. Relationships and Hierarchies

Ontologies are often hierarchical in nature. For a recipe ontology:

Ingredient may have subclasses like "Vegetable," "Meat," and "Spice."

Subcategories are cooking methods such as dry heat and moist heat.

Cuisine could be divided into "Italian," "Indian," "Mexican," etc.

These hierarchies allow for detailed categorization and inference.

4. Constraints and Rules

The ontology would then include subsequent rules such as:

Certain recipes, which demand special cooking or equipment.

Limiting ingredient combinations (such as dietary constraints).

Rules for categorizing recipes by cuisine or difficulty level.

5. Potential Applications This ontology could be used for:

Recipe recommendation systems.

Nutritional analysis tools. Automatic grocery list generation. Cuisine culture explored.

6. Suggestions for Improvement Improvements might be made by using the following

enhancements, based on a typical analysis of recipe ontologies: Interoperability means the ability

of being compatible with external databases or standards, such as Schema.org. Localization:

Support for multilingual labels for being applicable worldwide. Temporal information: Adding such

things as properties for seasonal ingredients, cooking times.

# Describe the conceptualization approach

**Conceptual Approach: Top-Down**

For the development of Recipe Ontology, we used the conceptualization approach by a top-down method.

From highly general and abstracted to narrow categorical specifics, the process steps were to achieve well-structured intuitive representations of this particular recipe domain without breaking any logical consistencies while making it practical in their use.

Step 1: Abstract Concept

The abstract, most general concept, used as a root class is **'Recipe'**. By keeping in mind that

fundamental concept, I was able to think about a few things that could be and must be associated with a recipe basically and always:

1. Ingredients

2. Preparation Steps

3. Cookware

4. Ways of Cooking

5. Timing

Step 2: Expanding to Broader Concept Classes

To specify each component of these elements and their relationships with recipes in a standard

manner, every single one was expanded by :

Ingredients - Included under broad heads like:

- Vegetables
- Proteins
- Spices
- Fats
- Liquids

Steps of Preparation: this constitutes significant processes like:

- Chopping
- Mixing
- Boiling
- Searing
- Roasting

Utensils or Cooking Equipment: The following utensil/equipment classifications were categorized:

- Pots
- Pans
- Knives

Each one explained in a very specific category as well as "Non-Stick Pans" and "Chef's Knives."

Cooking Methods: In the processes

- Grilling
- Saute
- Steam

these also explained how equipment and the stages of preparation go hand in glove.

Time needed: Classification According to the Duration:

- Prep time
- Cook Time

Step 3: Refining into Specific Details

These categories then broke down to their more specific elements so that the granularity and

comprehensiveness should be realized:

Ingredients- In this category fell "Tomatoes," "Chicken," "Cumin," "Butter," and "Olive Oil." Each was just an example that could be specific. - During Preparation Steps, for clarity of representation, adding "temperature," "duration," "tools

required," etc., made each step appropriate.

Results and Advantages

By the top-down approach:

1. I ensured that the ontology had a hierarchical structure from abstract to concrete.

2. The relations between ingredients, steps, tools, and methods were logically consistent with the

real world and how a kitchen really works.

3. This leads to a wide and user-friendly framework, aligning the ontology with established culinary

classifications and practical cooking processes.

# concept glossary and relationship glossary

concept glossary:

| Concept | Definition | Source | Synonyms |
|---|---|---|---|
| **Cooking_Method** | Moist heat cooking technique simply use by applying steam or liquid to cook food. Food cooked with steam is more healthy and extremely juicy and tender as compared to other methods. | 17 Types of Cooking Methods and Techniques in 2023 - Hinz Cooking | |
| **Grilling** | you give color to your food by exposing heat. | 17 Types of Cooking Methods and Techniques in 2023 - Hinz Cooking | |
| **Blanching** | Blanching is the most used cooking process to boil or scaled the vegetables in hot water and after a short duration, hot water needs to drain out and vegetable needs to place in cold or chilled water. | 17 Types of Cooking Methods and Techniques in 2023 - Hinz Cooking | |
| **Roasting** | In this cooking technique, food cooks initially on the high 'indirect heat' and after few minutes switch to 'medium heat' so the food cook from inside. | 17 Types of Cooking Methods and Techniques in 2023 - Hinz Cooking | |
| **Quick_Recipe** | a recipe that require 30 minutes or less | | easy recipe |

| | | | |
|---|---|---|---|
| **Slow-Cooked_Recipe** | a recipe that require between 30 minutes to 2 hours | | hard recipe |
| **Cookware** | Cookware is a basic necessity in any kitchen. It is very important to select the right cookware for your cooking needs. | https://cookwarespace.com/types-of-cookware/ | |
| **Anodized_Aluminum** | a good conductor of heat and it is light in weight & non-reactive | https://cookwarespace.com/types-of-cookware/ | |
| **Enameled_Cast_Iron** | type of cookware that has a enamel coating on the surface. The enamel coating is usually made from porcelain. | https://cookwarespace.com/types-of-cookware/ | |
| **Ingredient** | | ontology source :https://github.com/waleedahmed7/Recipe-Ontology/blob/main/Recipe%20Ontology.owl | |
| **Imperial** | relating to, befitting, or suggestive of an empire or an emperor , of or relating to the United Kingdom as distinguished from the constituent parts | Imperial Definition & Meaning - Merriam-Webster | |
| **Metric** | a mathematical function that associates a real nonnegative number analogous to distance with each pair of elements in a set such that the number is zero only if the two elements are identical | Metric Definition & Meaning - Merriam-Webster | |
| **Recipe_Source** | where the recipe came from | | |

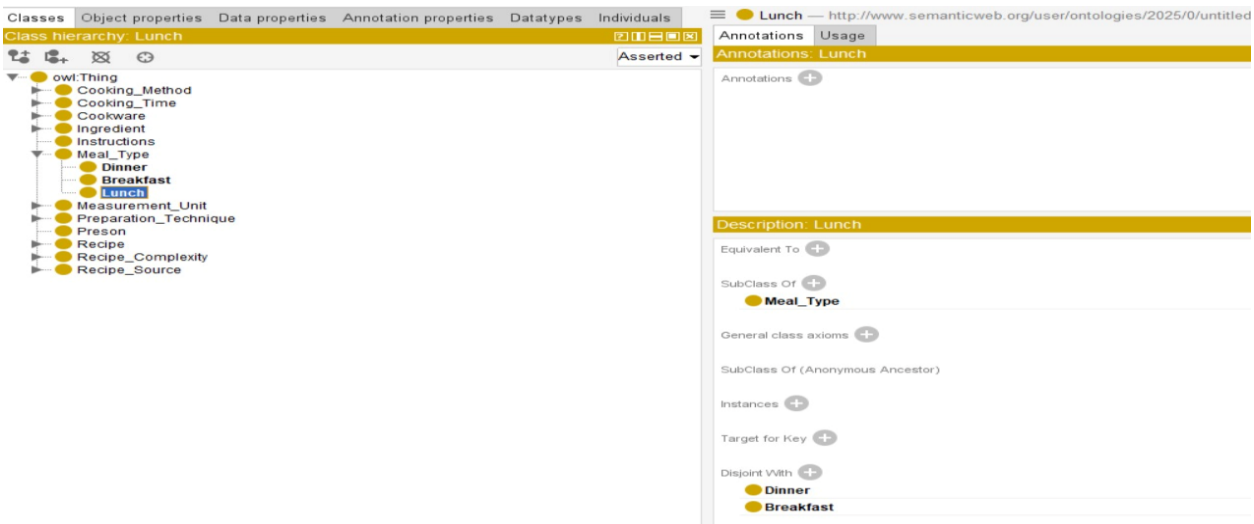
| | | | |
|---|---|---|---|
| **Chef_Recipe** | In Chef, recipes are sets of instructions that define the desired configuration of a system. | Writing Chef Recipes - Tutorial | |

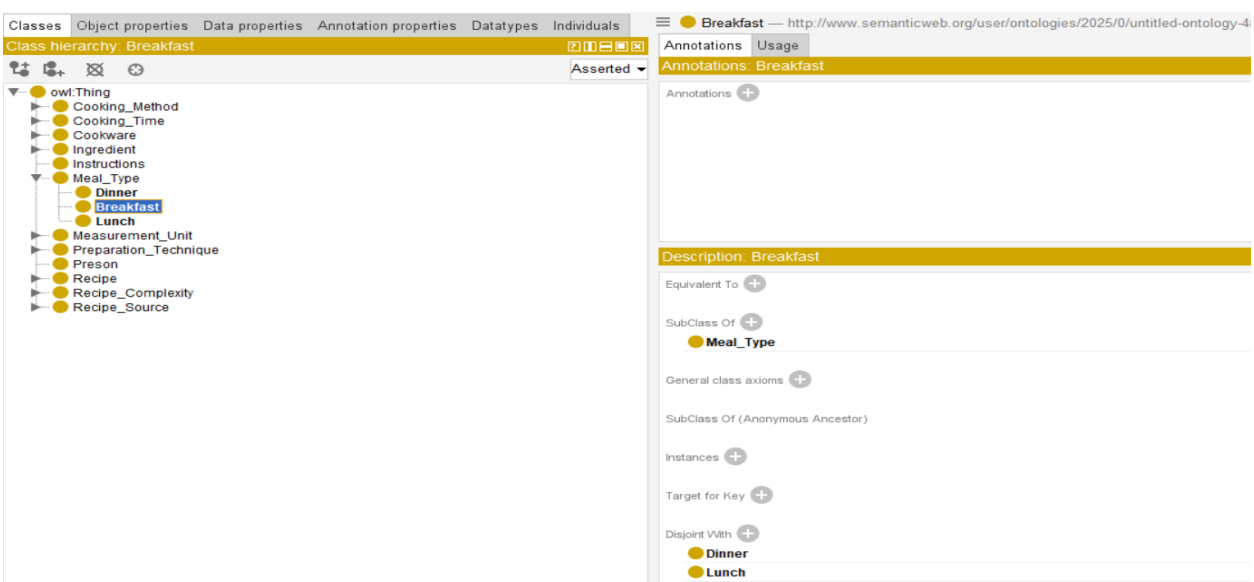
## Relationship Glossary :

| **Relationship** | **Definition** |
|---|---|
| **Author** | someone who write . Example: every recipe has a writer .<br>ontology source : https://github.com/acsalcedo/recipe-ontology-ws/blob/master/ontology/recipes.owl |
| **CookingRecipe** | the person who cooks . Example: every person cooks a recipe. |
| **Tool** | a tool used to make a recipe . Example: In every recipe we need a tool to use. |
| **wasDerivedFrom** | The source or the place where the recipe instruction is .Example: every recipe has a recipe source |
| **wasGivenFrom** | a person who give me the recipe . Example: every recipe was given from someone . |
| **hasPrepTime** | a time to make a recipe . Example: every recipe has a prepare time. |
| **cookTime** | how much time we need to make a recipe . Example: a recipe take 45 minutes . |

| | |
|---|---|
| **hasRecipeInstructions** | instruction to make a recipe . Example: this is the instruction for a recipe |
| **HasUnit** | Example: you need cup of sugar. |
| **HasValue** | Example: you need 3 grams of buttermilk. |

# Define Classes

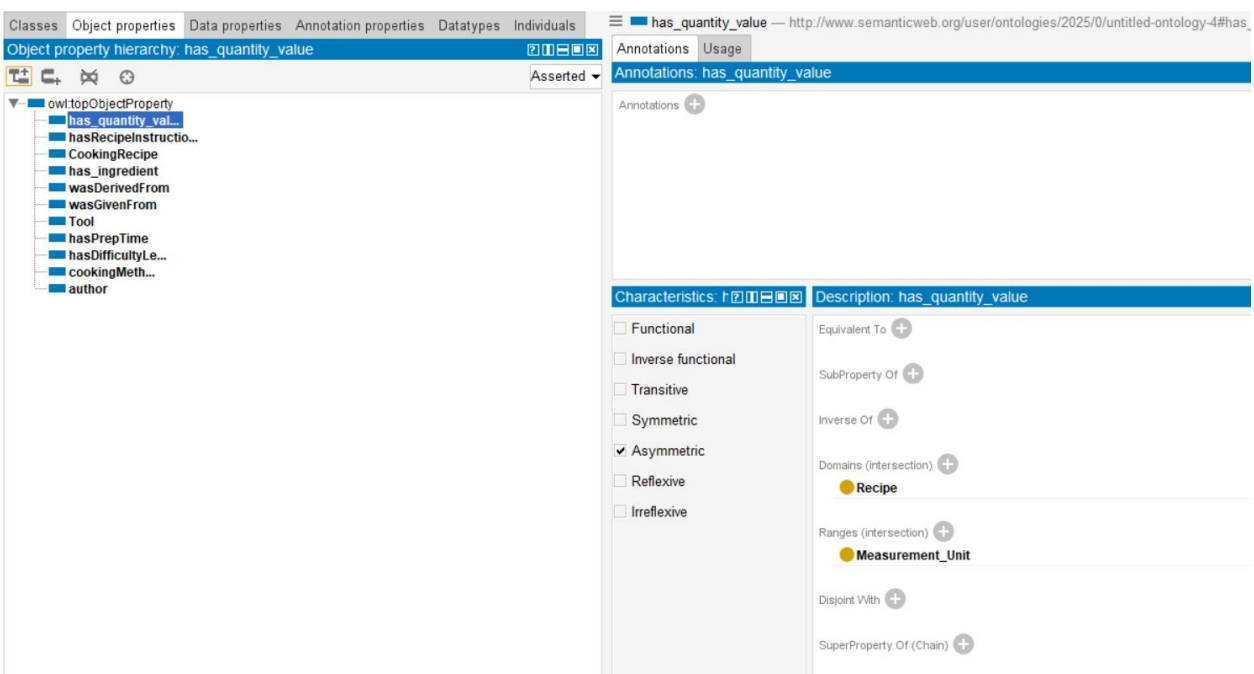


The picture demonstrates how "Lunch," "Dinner," and "Breakfast" are in fact separated. Breakfast , Lunch and  Dinner represent distinct meal types, occurring at different times of the day. The "Disjoint With" link between these classes serves as evidence for this. A clear and unambiguous representation of meal kinds in the ontology is ensured by the fact that an individual cannot be an instance of more than one of these classes. By effectively defining  disjoint axioms so ontology developers can create more accurate and informative knowledge representations.

## Define Object Properties and Data properties

These are screenshots of the object & data properties.

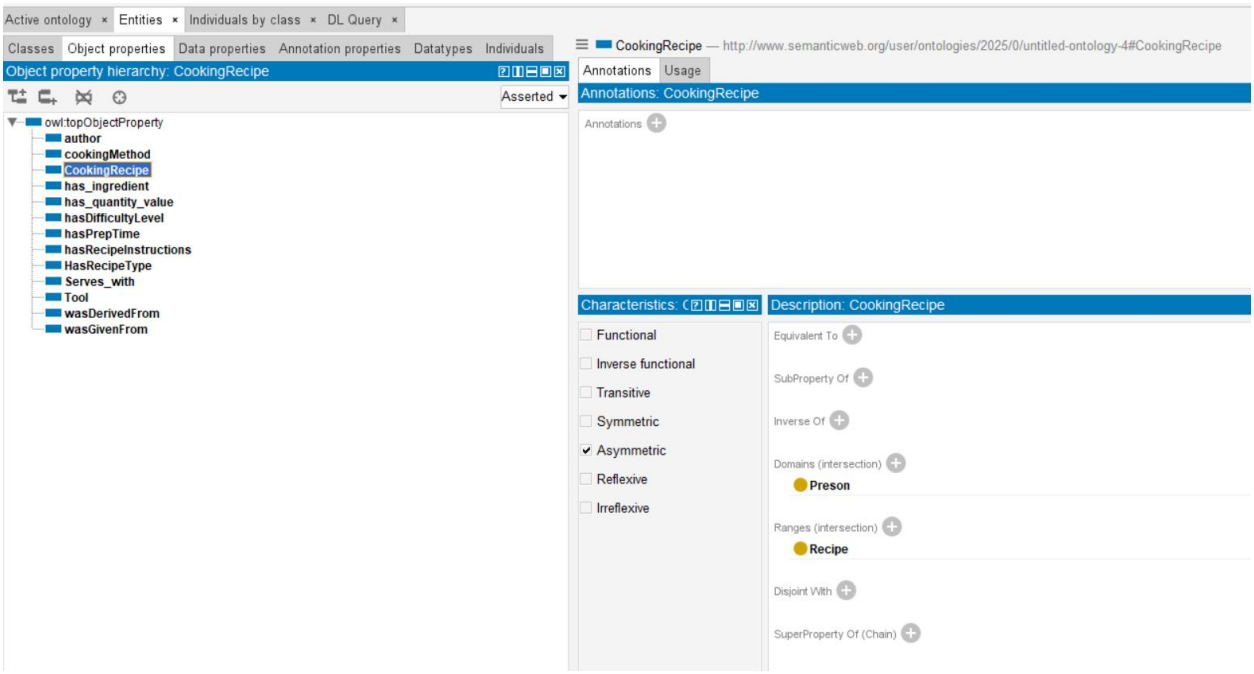



Because it depicts a directional relationship in which the quantity value is linked to the recipe or ingredient but not the other way around, the "has_quantity_value" is an object properties that relate two classes : Recipe & Measurement_Unit .

**Property:** has_quantity_value

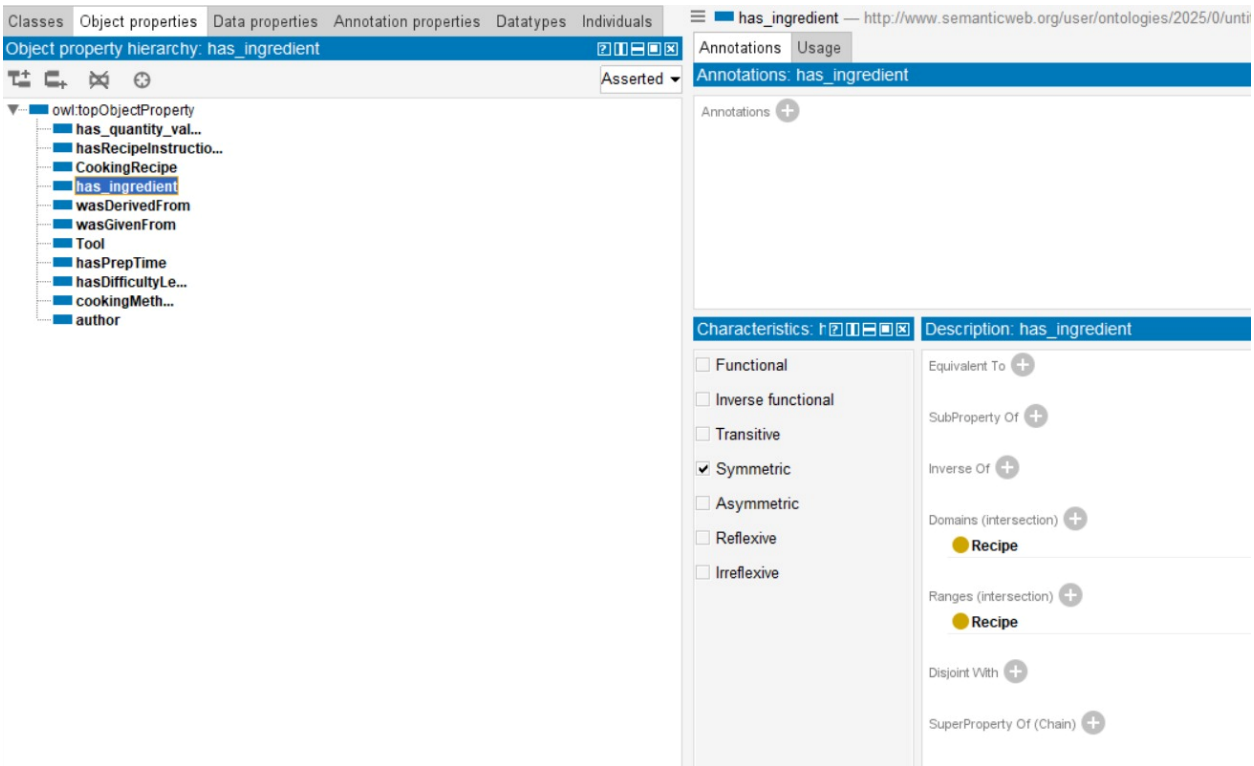
If Recipe has a  Measurement value (Recipe has_quantity_value  a Measurement_Unit) , then Measurement_Unit cannot have a quantity value Recipe (Measurement_Unit  has_quantity_value Recipe is False ).

Because it depicts a directional relationship that doesnot flow in both directions, the "has_quantity_value" trait is not symmetric.



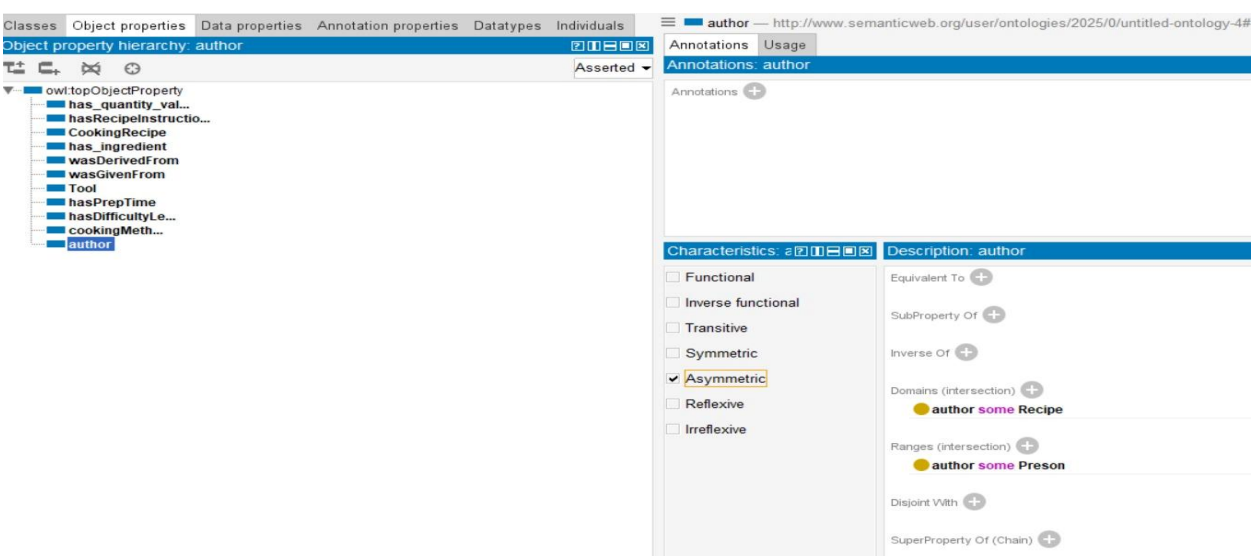

**Property:** CookingRecipe

If a Person is cooking a Recipe (Person CookingRecipe Recipe) , then Recipe  cannot be cooking a Person (Recipe  CookingRecipe Person is False ).

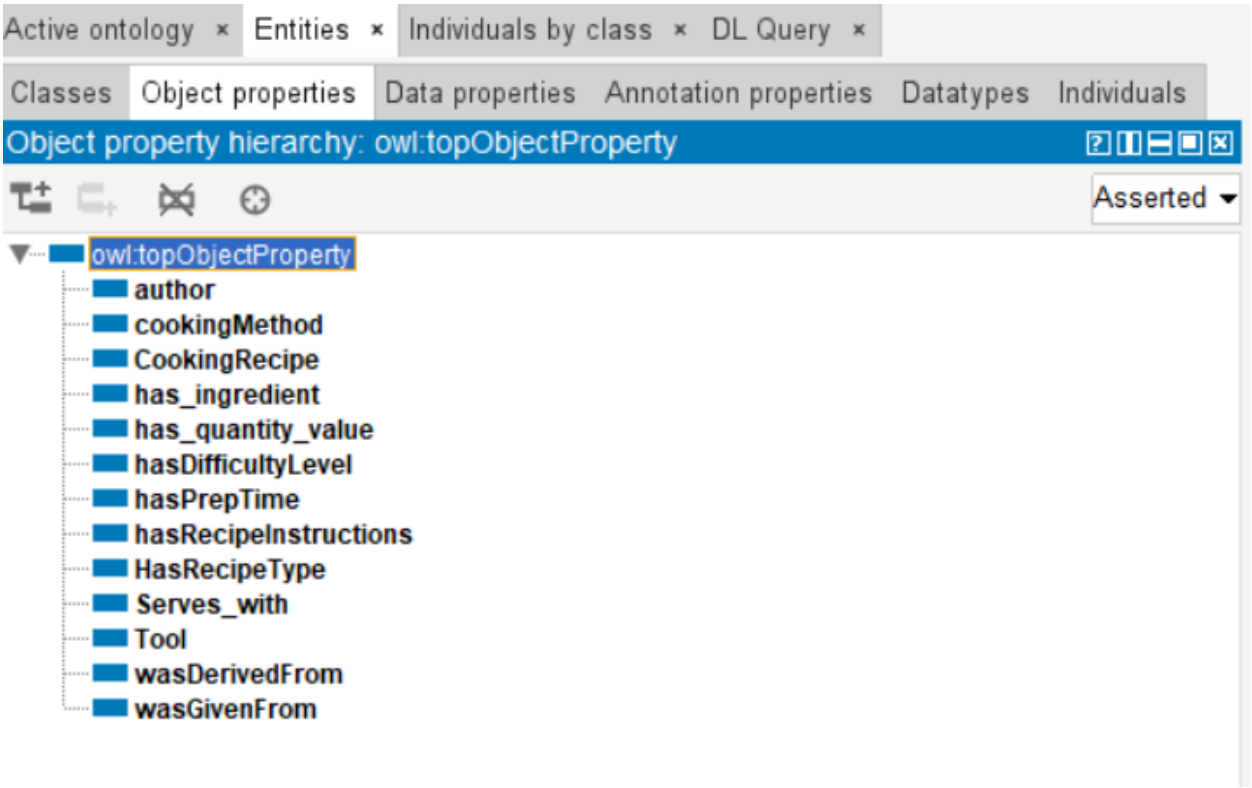



symmetric property is one which, if relation holds from one direction, then it must hold from the other direction.

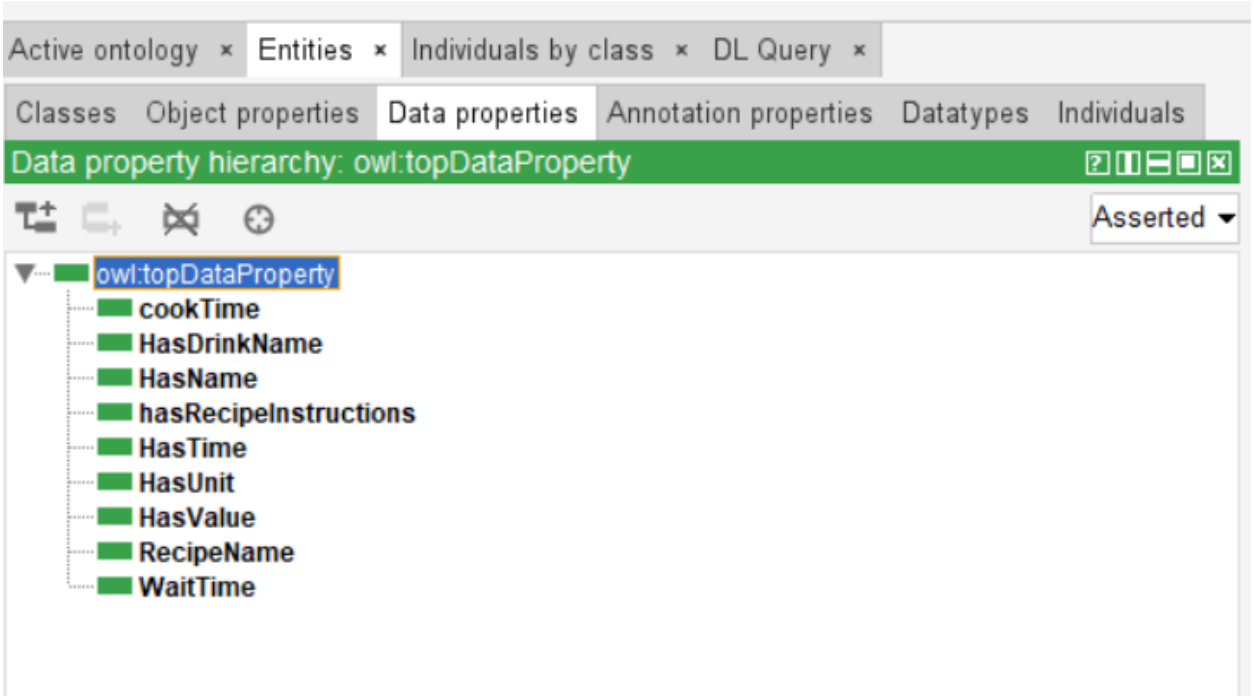
According to some interpretations, the "has_ingredient" property is symmetric, meaning that a recipe's connection with its components is reciprocal. But it's crucial to take into account the particular context and the ontology's intended modeling of the relationship.



Because it denotes a directional relationship in which the author develops and write the recipe rather than the other way around, the "author" characteristic is asymmetrical.




The picture depicts an ontology with a hierarchy of object properties that are connected to recipes. Features like "has_quantity_value," "has_ingredient," "author" and “Tool ” imply that its goal is to represent and make sense of recipes, ingredients, and their connections. Identifying the creator of a recipe or locating recipes with particular ingredients are just two applications for this ontology.




The image shows the hierarchy of data properties in an ontology. Data characteristics (e.g., "cookTime" or "RecipeName" of a recipe and "HasName" for a person) define individual qualities. It is clear from the property names that recipes are the main focus of this ontology. These data attributes can be used to store and query recipe specifics, such as names, cooking times, and associated instructions.

# Define Instances

## SPARQL Queries:

SPARQL Query Editor  About  Tables  Conductor  Facet Browser  Permalink

Extensions: cxml  save to dav  sponge  User: SPARQL

Default Data Set Name (Graph IRI)

http://dbpedia.org

Query Text

```
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?chef ?Name
FROM <http://dbpedia.org>
WHERE {
  ?chef rdf:type dbo:Chef .
  ?chef rdfs:label ?Name .
}
LIMIT 100
```

Results Format: HTML

Execute Query  Reset

Execution timeout: 30000 milliseconds

Options
- [x] Strict checking of void variables
- [x] Strict checking of variable names used in multiple clauses but not logically connected to each other
- [ ] Suppress errors on wrong geometries and errors on geometrical operators (failed operations will return NULL)

The SPARQL query shown above was executed to retrieve information about chefs from DBpedia. This query uses the `dbo:Chef` class to identify entities categorized as chefs. The

`?Name` variable is used to extract the labels of these entities. The query also includes a `LIMIT` clause to restrict the results to 100 entries.

Result of the above Query:

SPARQL | HTML5 table

| chef | Name |
|---|---|
| http://dbpedia.org/resource/Carla_Hall | "Carla Hall"@en |
| http://dbpedia.org/resource/Carla_Hall | "كارلا هال"@ar |
| http://dbpedia.org/resource/Carla_Hall | "Карла Халь"@uk |
| http://dbpedia.org/resource/Carme_Ruscalleda | "Carme Ruscalleda"@en |
| http://dbpedia.org/resource/Carme_Ruscalleda | "Carme Ruscalleda i Serra"@ca |
| http://dbpedia.org/resource/Carme_Ruscalleda | "Carme Ruscalleda"@de |
| http://dbpedia.org/resource/Carme_Ruscalleda | "Carme Ruscalleda"@es |
| http://dbpedia.org/resource/Carme_Ruscalleda | "Carme Ruscalleda"@fr |
| http://dbpedia.org/resource/Carme_Ruscalleda | "カルメ・ルスカイェーダ"@ja |
| http://dbpedia.org/resource/Carrie_Nahabedian | "Carrie Nahabedian"@en |
| http://dbpedia.org/resource/Carson_Gulley | "Carson Gulley"@en |
| http://dbpedia.org/resource/Cat_Cora | "Cat Cora"@en |
| http://dbpedia.org/resource/Cat_Cora | "كات كورا"@ar |
| http://dbpedia.org/resource/Cat_Cora | "キャット・コーラ"@ja |

The query successfully retrieved information about chefs such as Carla Hall, Carme Ruscalleda, and Cat Cora.

SPARQL Query Editor About Tables Conductor Facet Browser Permalin

Extensions: cxml save to dav sponge User: SPA

Default Data Set Name (Graph IRI)

http://dbpedia.org

Query Text

```
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?food ?foodName
WHERE {
  ?food rdf:type dbo:Food .
  ?food rdfs:label ?foodName .
}
LIMIT 100
```

Results Format HTML

Execute Query Reset

The SPARQL query shown above was executed to retrieve information about Food from DBpedia. This query uses the `dbo:Food` class to identify entities categorized as Food. The `?foodName` variable is used to extract the labels of these entities. The query also includes a `LIMIT` clause to restrict the results to 100 entries.

## Result of the above Query:

SPARQL | HTML5 table

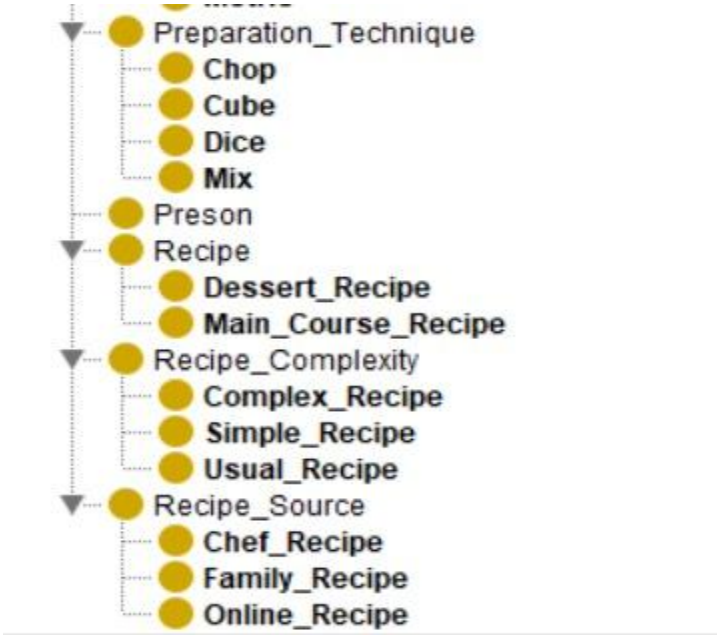
| food | foodName |
|---|---|
| http://dbpedia.org/resource/Cadbury_Clusters | "Cadbury Clusters"@en |
| http://dbpedia.org/resource/Cadbury_Creme_Egg | "Cadbury Creme Egg"@en |
| http://dbpedia.org/resource/Cadbury_Fingers | "Cadbury Fingers"@en |
| http://dbpedia.org/resource/Caesar_salad | "Caesar salad"@en |
| http://dbpedia.org/resource/Caffè_mocha | "Caffè mocha"@en |
| http://dbpedia.org/resource/Café_de_Paris_sauce | "Café de Paris sauce"@en |
| http://dbpedia.org/resource/Café_de_olla | "Café de olla"@en |
| http://dbpedia.org/resource/Calas_(food) | "Calas (food)"@en |
| http://dbpedia.org/resource/Caldo_gallego | "Caldo gallego"@en |
| http://dbpedia.org/resource/California_roll | "California roll"@en |
| http://dbpedia.org/resource/Camaron_rebosado | "Camaron rebosado"@en |
| http://dbpedia.org/resource/Campa_Cola | "Campa Cola"@en |
| http://dbpedia.org/resource/Campari | "Campari"@en |
| http://dbpedia.org/resource/Canal+_Gol_(Poland) | "Canal+ Gol (Poland)"@en |

SPARQL | HTML5 table

| | |
|---|---|
| http://dbpedia.org/resource/Bayram_(Turkey) | "Bayram (Turkey)"@en |
| http://dbpedia.org/resource/Bazaar_Canton | "Bazaar Canton"@en |
| http://dbpedia.org/resource/Bazin_(bread) | "Bazin (bread)"@en |
| http://dbpedia.org/resource/Bazlama | "Bazlama"@en |
| http://dbpedia.org/resource/Beach_ball | "Beach ball"@en |
| http://dbpedia.org/resource/Beach_pizza | "Beach pizza"@en |
| http://dbpedia.org/resource/Bean_pie | "Bean pie"@en |
| http://dbpedia.org/resource/Bean_salad | "Bean salad"@en |
| http://dbpedia.org/resource/Bean_sprouts_chicken | "Bean sprouts chicken"@en |
| http://dbpedia.org/resource/Bear_claw | "Bear claw"@en |
| http://dbpedia.org/resource/Beaten_biscuit | "Beaten biscuit"@en |
| http://dbpedia.org/resource/Beauty | "Beauty"@en |
| http://dbpedia.org/resource/Bebinca | "Bebinca"@en |
| http://dbpedia.org/resource/Becherovka | "Becherovka"@en |
| http://dbpedia.org/resource/Bed_warmer | "Bed warmer"@en |

The query successfully retrieved information about Foods such as Cadbury Clusters, Calas, and California Roll.

## Ontology Class Hierarchies:

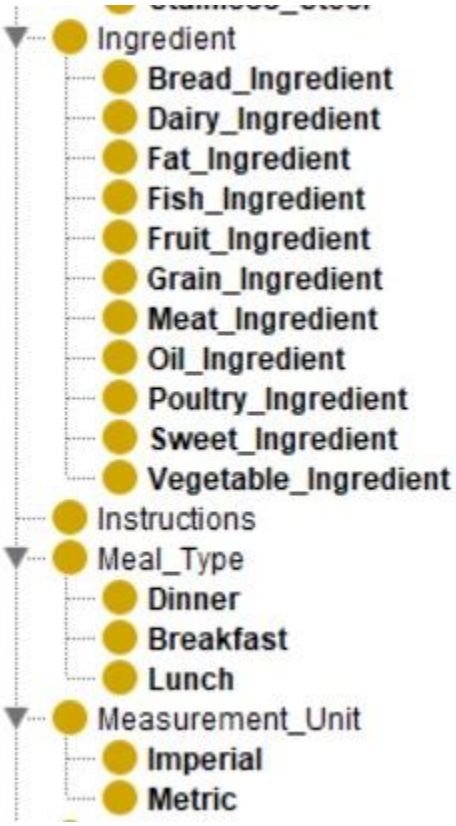


This image represents an ontology's class hierarchy, showing different entities related to recipes.

It includes parent classes (e.g., Recipe) and their subclasses (e.g., Dessert_Recipe, Main_Course_Recipe).

The hierarchy also shows other related concepts such as Preparation_Technique and Recipe_Source.

Recipes are linked to related concepts like Preparation_Technique, Recipe_Source, and Ingredient by this ontology, which also arranges them into hierarchical categories (e.g., Dessert_Recipe, Main_Course_Recipe). Recipes may be easily categorized and retrieved according to their kind, ingredients, preparation techniques, and sources. It is a helpful framework for creating recipes and meal planning since it also allows for nutritional filtering, cooking time classifications, and serving size variations.

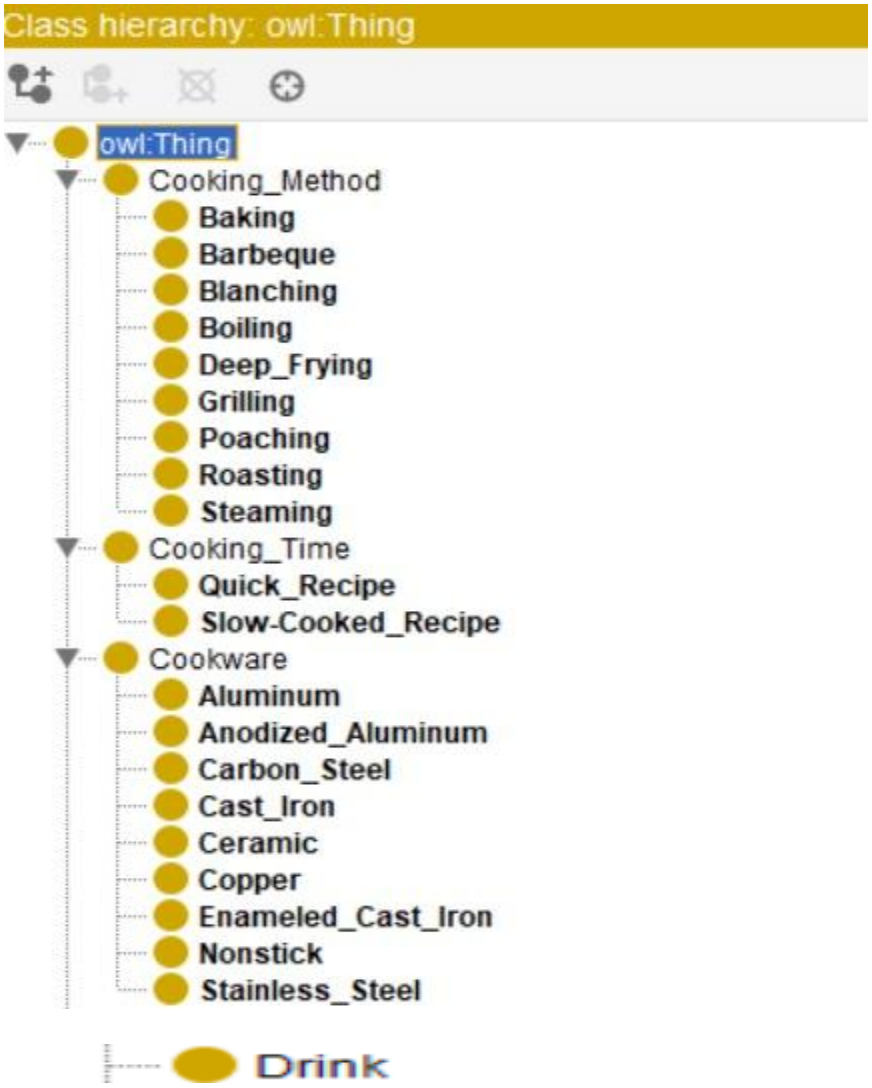


This image displays the hierarchy of ingredients and related entities within the ontology.

The Ingredient class is the parent entity, which branches into various specific ingredient categories:

Bread_Ingredient, Dairy_Ingredient, etc., represent different types of ingredients used in recipes.

Many other entities relating to food are built on top of the Ingredient class. In order to facilitate their organization, accessibility, and utilization in recipe development and culinary situations, each subclass classifies ingredients according to their attributes.



This image focuses on methods, time, and tools related to cooking.

The parent class owl:Thing includes:

Cooking_Method: Lists different cooking techniques, such as Baking, Boiling, Grilling, and Poaching.

Cooking_Time: Classifies recipes based on preparation time, e.g., Quick_Recipe and Slow-Cooked_Recipe.

Cookware: Enumerates tools used in cooking, such as Aluminum, Cast_Iron, and Nonstick cookware.

By classifying and connecting many elements of the cooking process, this ontology facilitates recipe selection, meal planning, and kitchen management by grouping cooking techniques, time needs, and materials into structured categories.

## A_box:

- Beach_pizza: Recipe
- Bean_Salad: Recipe
- Cadbury_Clusters: Dessert_Recipe
- Cadbury_Creme_Egg: Sweet_Ingredient
- Café_de_olla: Drink
- Campa_Cola: Drink
- Carla_Hall: Person
- Carme_Ruscalleda: Person
- Карла_Халь: Person
- カルメ・ルスカイェーダ： Person
- (Campa_Cola, Cadbury_Clusters): Serves_with
- (Café_de_olla, Cadbury_Clusters): Serves_with
- (Carla_Hall): HasName
- (Carme_Ruscalleda): HasName